<u>Important Points – Unit 1</u>

- Quantum computing arises from the union of **information theory** and **quantum mechanics**.
- Information theory allowed computer science and communications to abstract away the **physical world**.
- A **Turing machine** is a model of computation based on **classical mechanics**.
- **Charles Bennett** and **Gilles Brassard** introduced a secure quantum protocol based on quantum **measurement**.
- Certain quantum phenomena could not be efficiently simulated by a **classical Turing machine**, as noted by **Richard Feynman** and **Yuri Manin**.
- David **Deutsch** defined a model of a **quantum Turing machine**.
- The **quantum circuit model** describes computation in terms of **quantum gates**.
- **Peter Shor's algorithm** showed that a quantum computer could factor integers in **polynomial time**.
- The security of the **RSA** cryptosystem relies on the difficulty of **integer factorization**.
- Quantum systems are fragile because of **decoherence**, caused by **environmental interactions**.
- The **no-cloning principle** states that an unknown quantum state cannot be reliably **copied**.
- **Grover's algorithm** provides a quadratic speedup for **unstructured search problems**.
- Simulation of **quantum systems** is a major application of **quantum computation**.
- Quantum information processing includes **computing, cryptography, communications, and games**.
- Despite infinite possible states of a qubit, measurement extracts only **one bit** of classical information.
- A **qubit** is the quantum analogue of a **classical bit**.
- Photon **polarization** is one physical realization of a **qubit**.
- A **polaroid filter** allows photons aligned with its **preferred axis** to pass through.
- Two orthogonal polaroids (horizontal + vertical) block **all light**.
- Inserting a **third polaroid** at 45° allows **some light** to pass.
- In Dirac notation, $\mathbf{|0\rangle}$ and $\mathbf{|1\rangle}$ form the **computational basis**.
- A general qubit can be written as $\mathbf{a|0\rangle + b|1\rangle}$, where $\mathbf{|a|^2 + |b|^2 = 1}$.
- A qubit with both coefficients non-zero is said to be in **superposition**.
- The probability of passing through a polaroid is the **square of the amplitude** in the polaroid's axis direction.
- The **inner product** $\langle v|v\rangle$ for a state vector is always equal to **1**.
- Two vectors are orthogonal if their inner product equals **0**.
- An **orthonormal basis** consists of vectors that are **unit length** and mutually **orthogonal**.
- In photon polarization, $|\uparrow\rangle$ represents **vertical polarization** and $|\rightarrow\rangle$ represents **horizontal polarization**.
- Measuring a state $\mathbf{a|0\rangle + b|1\rangle}$ in the standard basis yields $|0\rangle$ with probability $\mathbf{|a|^2}$ and $|1\rangle$ with probability $\mathbf{|b|^2}$.
- Measurement transforms the quantum state into one of the **basis states**.

- After measurement, repeating the measurement in the same basis yields the **same result** with probability **1**.
- A superposition is **basis-dependent**; a state may be a superposition in one basis but not in another.
- The **Hadamard basis** consists of states $|\nearrow\rangle$ and $|\nwarrow\rangle$.
- Measuring $|\nearrow\rangle$ in the Hadamard basis is **deterministic**, while measuring it in the standard basis is **probabilistic**.
- Distinct superpositions like $(1/\sqrt{2})(|0\rangle+|1\rangle)$ and $(1/\sqrt{2})(|0\rangle-|1\rangle)$ represent **different states**.
- Because of measurement collapse, it is impossible to determine the **original state** of a qubit from repeated measurements.
- A qubit cannot be **cloned**, due to the **no-cloning theorem**.
- Even though a qubit can be in infinitely many states, only **one bit** of classical information can be extracted from it.
- Systems with three- or n-dimensional vector spaces are called **qutrits** and **qudits**, but they are computationally equivalent to **qubits**

1. A photon that passes through a polaroid becomes _____ polarized in the direction of the polaroid's axis.
   Answer: aligned
2. A quantum bit, or _____, has a continuum of possible values.
   Answer: qubit
3. Any state represented by a unit vector _____$|0\rangle$ + _____$|1\rangle$ is a legitimate qubit value.
   Answer: a, b
4. The amplitudes in a qubit state can be _____ numbers, not just real numbers.
   Answer: complex
5. A two-state quantum system can be represented as a _____ combination of two basis states.
   Answer: linear
6. In quantum mechanics, the standard notation for states is called _____ notation.
   Answer: Dirac's bra–ket
7. The two distinguished states for qubits, $|0\rangle$ and $|1\rangle$, must be _____.
   Answer: orthonormal
8. The length or norm of a quantum state vector $|v\rangle$ is always equal to _____.
   Answer: 1
9. In the standard basis, $|0\rangle$ is represented as $(1,0)^T$ and $|1\rangle$ as _____.
   Answer: $(0,1)^T$
10. A qubit can take not only the values $|0\rangle$ and $|1\rangle$ but also any _____ of them.
    Answer: superposition
11. A basis is orthonormal if all its elements are of unit length and mutually _____.
    Answer: orthogonal
12. The inner product of a bra and a ket is sometimes called a _____.
    Answer: bracket
13. A measurement device must have two preferred states forming an _____ basis.
    Answer: orthonormal

14. Measuring a state $|v\rangle = a|u\rangle + b|u\perp\rangle$ yields $|u\rangle$ with probability _____.
    Answer: $|a|^2$
15. After measurement, the qubit's state collapses to one of the _____ states of the basis.
    Answer: eigen
16. The probabilistic outcome of measurement is a fundamental _____ of quantum mechanics.
    Answer: axiom
17. Even though a qubit can be in infinitely many states, measurement yields at most one _____ bit of information.
    Answer: classical
18. The earliest quantum key distribution protocol is called _____.
    Answer: BB84
19. In the BB84 protocol, qubits can be encoded in either the standard basis or the _____ basis.
    Answer: Hadamard
20. In the standard basis, bit 0 is encoded as $|\uparrow\rangle$ and bit 1 as _____.
    Answer: $|\rightarrow\rangle$
21. In the Hadamard basis, 0 is encoded as $|\nearrow\rangle = (|\uparrow\rangle + |\rightarrow\rangle)/\sqrt{2}$, and 1 as _____.
    Answer: $|\nwarrow\rangle = (|\uparrow\rangle - |\rightarrow\rangle)/\sqrt{2}$
22. An eavesdropper who measures qubits in the wrong basis introduces a detectable _____ rate.
    Answer: error
23. The impossibility of reliably copying an unknown quantum state is called the _____ principle.
    Answer: no-cloning
24. Two vectors $|v1\rangle$ and $|v2\rangle$ are orthogonal if their inner product $\langle v1|v2\rangle$ = _____.
    Answer: 0
25. Multiplying a state vector by a modulus-one complex number only changes its _____ phase, not the state itself.
    Answer: global

1. In quantum mechanics, multiple-qubit state spaces are formed using the _____ product of vector spaces.
   Answer: tensor
2. For two qubits, the state space dimension is _____.
   Answer: 4
3. The computational basis for a 3-qubit system consists of _____ basis states.
   Answer: 8

6. A state that cannot be written as a product of single-qubit states is called an _____ state.
   Answer: entangled
7. The four maximally entangled 2-qubit states are known as the _____ states.
   Answer: Bell

11. The probability of an outcome in quantum measurement is given by the _____ of the amplitude.
    Answer: squared magnitude
12. Measuring only part of a system leaves the unmeasured qubits in a _____ state.
    Answer: conditional

13. Measurement is always _____ and changes the quantum state.
    Answer: probabilistic
14. Entangled states like Bell pairs can also be measured in the _____ basis.
    Answer: Bell

16. The entanglement-based QKD protocol was proposed by _____ in 1991.
    Answer: Artur Ekert
17. Ekert's protocol uses pairs of _____ qubits shared between Alice and Bob.
    Answer: entangled
18. Security in entanglement-based QKD is checked using violation of _____ inequality.
    Answer: Bell's
19. Any attempt by Eve to measure or clone the qubits breaks the _____ and introduces errors.
    Answer: entanglement
20. Unlike BB84, Ekert's protocol bases its security on quantum _____ rather than computational hardness.
    Answer: physics

1. **Who introduced the concept of a quantum Turing machine?**
    - **Answer:** David Deutsch
2. **What is the fundamental unit of quantum computation?**
    - **Answer:** Qubit
3. **Which two theories combine to form quantum information theory?**
    - **Answer:** Information theory and quantum mechanics
4. **Who proposed the first quantum key distribution protocol (BB84)?**
    - **Answer:** Charles Bennett and Gilles Brassard
5. **Which algorithm provides a quadratic speedup for unstructured search?**
    - **Answer:** Grover's Algorithm
6. **In which year did Peter Shor publish his integer factorization algorithm?**
    - **Answer:** 1994
7. **What principle states that quantum states cannot be perfectly copied?**
    - **Answer:** No-cloning principle
8. **What does $|\uparrow\rangle$ represent in photon polarization?**
    - **Answer:** Vertical polarization
9. **Write the condition for a qubit state $a|0\rangle+b|1\rangle$ to be valid.**
    - **Answer:** $|a|^2 + |b|^2 = 1$
10. **What is the probability of obtaining $|0\rangle$ when measuring the state $a|0\rangle+b|1\rangle$?**
    - **Answer:** $|a|^2$

1. **Differentiate between a classical bit and a qubit.**
    - **Answer:**
        - A classical bit can take values 0 or 1.
        - A qubit can exist in $|0\rangle$, $|1\rangle$, or any superposition $a|0\rangle+b|1\rangle$, where $|a|^2+|b|^2=1$.
2. **Explain why inserting a third polaroid at 45° between two orthogonal polaroids allows light to pass.**
    - **Answer:**
        - Quantum mechanics models polarization as superpositions.
        - The intermediate polaroid projects photons into a new basis, giving them nonzero amplitude to pass through the last filter.
3. **What happens to a qubit state when it is measured?**
    - **Answer:**
        - Measurement collapses the state into one of the basis states of the measuring device.
        - The outcome is probabilistic, with probabilities determined by the square of amplitudes.
4. **State the difference between global phase and relative phase in qubit states.**
    - **Answer:**
        - Global phase ($e^{i\varphi}$) has no physical meaning and does not affect the state.
        - Relative phase between components affects interference and measurement outcomes.
5. **Why can only one bit of classical information be extracted from a qubit?**
    - **Answer:**
        - Measurement collapses a qubit to one of two basis states.
        - Thus, a single measurement yields only one classical bit, regardless of the infinite possible qubit states.
6. **What is meant by an orthonormal basis in quantum mechanics?**
    - **Answer:**
        - A set of vectors that are unit length and mutually orthogonal.
        - Example: $\{|0\rangle, |1\rangle\}$.
7. **Mention two physical realizations of a qubit.**
    - **Answer:**
        - Photon polarization
        - Electron spin
8. **State one limitation of quantum computers highlighted in early research.**
    - **Answer:**
    - Quantum states are fragile and easily disturbed by decoherence, making error correction challenging.

1. **Q:** What is the basic unit of quantum information?
   A: Qubit.
2. **Q:** In Dirac notation, what symbol represents a state vector?
   A: A ket $|v\rangle$.
3. **Q:** What condition must $|0\rangle$ and $|1\rangle$ satisfy in a qubit system?
   A: They must be orthonormal.
4. **Q:** What is the norm of any valid quantum state vector?
   A: 1.
5. **Q:** What does the no-cloning principle state?
   A: An unknown quantum state cannot be copied reliably.
6. **Q:** What are the two common bases used in BB84 protocol?
   A: Standard basis and Hadamard basis.
7. **Q:** What is the probability that a qubit collapses to $|u\rangle$ when measured if its amplitude is *a*?
   A: $|a|^2$.
8. **Q:** What is the global phase of a quantum state?
   A: A multiplicative complex factor of modulus one that has no physical meaning.
9. **Q:** Name one physical realization of a qubit.
   A: Photon polarization (others: electron spin, atomic energy levels).
10. **Q:** What is the earliest quantum key distribution protocol called?
    A: BB84.

1. **Q:** Differentiate between a bit and a qubit.
   A: A bit can take values 0 or 1, while a qubit can be in $|0\rangle$, $|1\rangle$, or any superposition $a|0\rangle + b|1\rangle$ with $|a|^2 + |b|^2 = 1$.
2. **Q:** Why can only one classical bit of information be extracted from a qubit?
   A: Because measurement yields one of two basis states, and measurement collapses the state, preventing further extraction.
3. **Q:** How does Eve's eavesdropping introduce errors in BB84 protocol?
   A: Measuring in the wrong basis changes the photon's polarization, causing about 25% error in Bob's measurements.
4. **Q:** Explain the difference between relative phase and global phase.
   A: Relative phase affects quantum interference and measurement outcomes, while global phase does not change the state physically.
5. **Q:** What does measurement of a qubit in the wrong basis imply?
   A: The result is probabilistic, and the qubit state collapses, possibly changing the outcome.
6. **Q:** Why is the Hadamard basis used in BB84 protocol?
   A: To introduce non-orthogonal states, making eavesdropping detectable.
7. **Q:** What is meant by span(S) in vector space terms?
   A: The set of all linear combinations of vectors in the set S.

1. **Q:** What is the dimension of the state space of 3 qubits?
   A: 23=82^3 = 823=8.
2. **Q:** Which mathematical operation combines quantum state spaces of subsystems?
   A: Tensor product.
3. **Q:** Name the four maximally entangled 2-qubit states.
   A: Bell states.

4. Q: Write the GHZ state for 3 qubits.
A: $|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$.

1. **Q:** In measurement, the probability of an outcome is the _____ of the amplitude.
   A: Squared magnitude.

2. What happens when only part of a multi-qubit system is measured?
   A: The measured qubits collapse to an outcome, while the unmeasured qubits collapse into a conditional state based on the outcome.

3. **Q:** Outline the main steps in Ekert's entanglement-based QKD protocol.
   Share entangled pairs between Alice and Bob.
   1. Measure qubits in randomly chosen bases.
   2. Keep only results where bases matched.
   3. Check correlations using Bell's inequality.
   4. Generate a secure key if no eavesdropping is detected.

Two-Mark Questions (with Answers)

1. **Q:** Explain why the state space of an n-qubit system has dimension 2^n.
   A: Each qubit has 2 possible basis states ($|0\rangle, |1\rangle$). For n qubits, the combined basis is all binary strings of length n, giving 2^n basis states.

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2. **Q:** Define an entangled state with an example.
   A: An entangled state is one that cannot be factored into product states of individual qubits. Example: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

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3. **Q:** Differentiate between product states and entangled states.
   A: Product states can be written as tensor products of individual qubits (e.g., $|0\rangle \otimes (|0\rangle + |1\rangle)/\sqrt{2}$), while entangled states cannot be factored (e.g., Bell states).

5. **Q:** Outline the main steps in Ekert's entanglement-based QKD protocol.
   A:
   - Share entangled pairs between Alice and Bob.
   - Measure qubits in randomly chosen bases.
   - Keep only results where bases matched.
   - Check correlations using Bell's inequality.
   - Generate a secure key if no eavesdropping is detected.